

KG COLLEGE OF ARTS AND SCIENCE
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JAVA PROJECT REPORT

Smart To-Do Task Manager with Priority Scheduli

Submitted by

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DECLARATION

I hereby declare that the project entitled "Smart To-Do Task Manager with Priority Scheduling**"** submitted in partial fulfillment of the requirements for the course Java Programming is a **record of my original work** carried out under the guidance of **Dr. Boopalan Sundaramoorthy,** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

Sudalai Balakumaran.p

CERTIFICATE

This is to certify that the project entitled **"Smart To-Do Task Manager with Priority Scheduling"** is a **bona fide work** carried out by **SUDALAIBALAKUMARAN, 1st Bsc CS 'A'**, B.Sc.Computer Science, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: KG College of Arts and Science

Date: 10.4.2026

Submitted for the Viva-Voce Examination held on

_____10.4.2026_____

Guide: **Head of the Department**(Technical Trainer, KGCAS)
(Technical Head, KGCAS)

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ABSTRACT

The Smart To-Do Task Manager is a JavaFX-based application designed to help users manage their daily tasks efficiently. The system allows users to add tasks with different priority levels such as High, Medium, and Low.

The application automatically sorts tasks based on priority, ensuring that important tasks are displayed first. Users can also delete completed or unwanted tasks.

This project demonstrates the use of Java programming concepts such as object-oriented programming, event handling, collections (ArrayList/ObservableList), and GUI development using JavaFX.

1. INTRODUCTION

Managing daily tasks is important for productivity. Many people use manual methods like notebooks, which are not efficient.

This system provides a digital solution where users can:

Add tasks

Assign priority

View sorted tasks

Delete tasks

The system is simple, user-friendly, and helps in better task organization.

1.1 Objective

- To develop a task management system using JavaFX
- To organize tasks based on priority
- To improve productivity and time management
- To apply Java concepts like OOP and GUI

1.2 Overview

The system is a GUI-based application where users can:

Enter task name

Select priority (High, Medium, Low)

View tasks in sorted order

Delete selected tasks

All tasks are stored temporarily using ObservableList.

1.3 Features

Add Task

Set Priority

Auto Sorting (High → Low

2. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

The analysis phase focuses on identifying the current issues in the existing system and proposing a computerized solution that improves efficiency and accuracy.

2.1 Problem Definition

Manual task tracking leads to:

Poor organization

Missing important tasks

No prioritization

2.2 Project Overview

The system provides:

Task input

Priority selection

Automatic sorting

Task deletion

2.3 Scopes and Limitation

Scope:

Personal task management

Student use

Simple productivity tool

Limitations:

No database (temporary storage)

Basic UI

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

3.1 Modules

Input Module – Enter task and priority

Processing Module – Sort tasks using priority

Storage Module – Stores tasks using ObservableList

Output Module – Displays tasks

Delete Module – Removes selected task

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Smart To-Do Task Manager with Priority Scheduling,** a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis• System Design• Program Development• Testing and Validation• Result and Documentation

4.1 Problem Analysis

Need for better task organization and prioritization.

4.2 System Design

Designed using JavaFX with GUI components.

4.3 Program Development

Created UI using JavaFX

Implemented event handling

Added sorting logic using Comparator

4.4 Testing

Tested with:

Empty input

Different priorities

Delete function

4.5 Result

System works correctly and sorts tasks automatically.

4.6. Algorithm

Algorithm for the Smart To-Do Task Manager with Priority Schedule

Start

Enter task name

Select priority

If input is valid

    Add task to list

    Sort list by priority

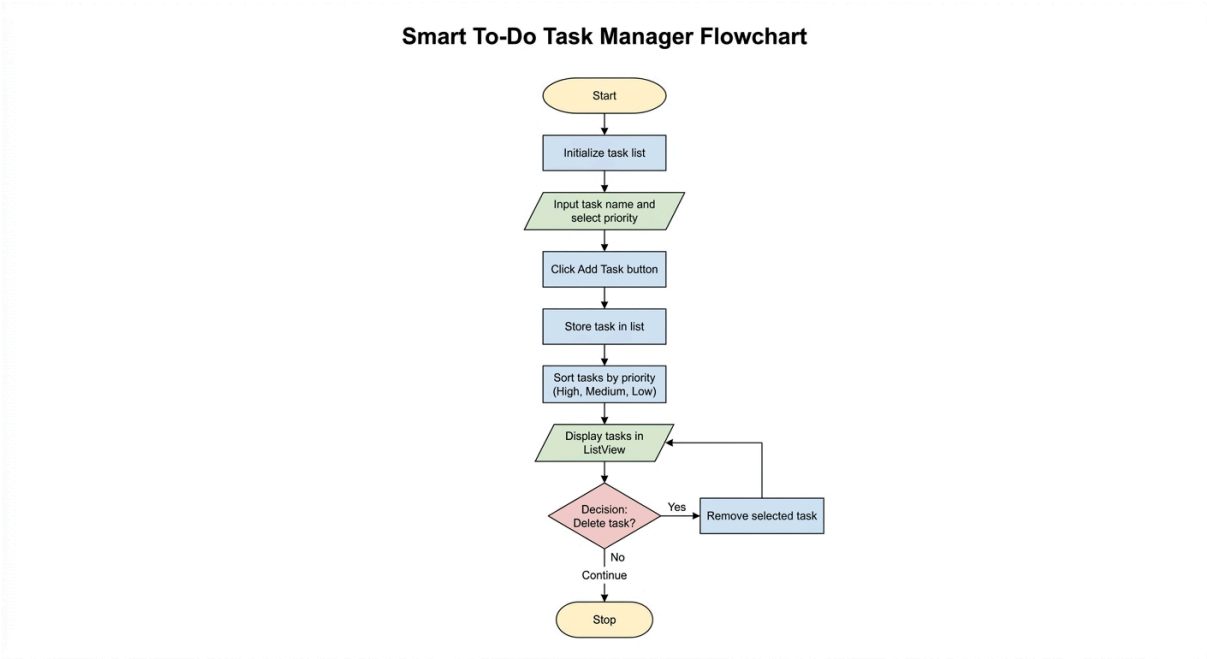
Display tasks

If delete selected

    Remove task

End

4.7. Flowchart



5. SOURCE CODE

```
import javafx.application.Application;
import javafx.collections.FXCollections;
import javafx.collections.ObservableList;
import javafx.geometry.Insets;
import javafx.scene.Scene;
import javafx.scene.control.*;
import javafx.scene.layout.*;
import javafx.stage.Stage;

import java.util.Comparator;

public class App extends Application {

    //      Task class
    static class Task {
        String name;
        String priority;
```

```
    public Task(String name, String priority) {
        this.name = name;
        this.priority = priority;
    }

    public int getPriorityValue() {
        switch (priority) {
            case "High": return 1;
            case "Medium": return 2;
            default: return 3;
        }
    }

    @Override
    public String toString() {
        return name + " [" + priority + "]";
    }
}

ObservableList<Task> taskList = FXCollections.observableArrayList();

ListView<Task> listView = new ListView<>(taskList);

@Override
public void start(Stage stage) {
```

```
//      Title
Label title = new Label("Smart To-Do Task Manager");
title.setStyle("-fx-font-size: 18px; -fx-font-weight: bold;");

//      Input
TextField taskField = new TextField();
taskField.setPromptText("Enter Task");

ComboBox<String> priorityBox = new ComboBox<>();
priorityBox.getItems().addAll("High", "Medium", "Low");
priorityBox.setPromptText("Select Priority");

Button addBtn = new Button("Add Task");

//      Add Task Logic
addBtn.setOnAction(e -> {
    String taskName = taskField.getText();
    String priority = priorityBox.getValue();

    if (!taskName.isEmpty() && priority != null) {
        taskList.add(new Task(taskName, priority));

        //      Sort by priority automatically
        FXCollections.sort(taskList,
```

```
Comparator.comparingInt(Task::getPriorityValue));

            taskField.clear();
            priorityBox.setValue(null);
        }
    });

    //      Delete Task
    Button deleteBtn = new Button("Delete Selected Task");
    deleteBtn.setOnAction(e -> {
        Task selected = listView.getSelectionModel().getSelectedItem();

        if (selected != null) {
            taskList.remove(selected);
        }
    });

    //      Layout
    VBox inputBox = new VBox(10,
            new Label("Add Task"),
            taskField,
            priorityBox,
            addBtn,
            deleteBtn
    );
```

```
    inputBox.setPadding(new Insets(10));
    inputBox.setStyle("-fx-border-color: black;");

    VBox listBox = new VBox(10,
            new Label("Task List (Auto Sorted by Priority)"),
            listView
    );

    listBox.setPadding(new Insets(10));
    listBox.setStyle("-fx-border-color: black;");

    HBox mainLayout = new HBox(20, inputBox, listBox);
    mainLayout.setPadding(new Insets(10));

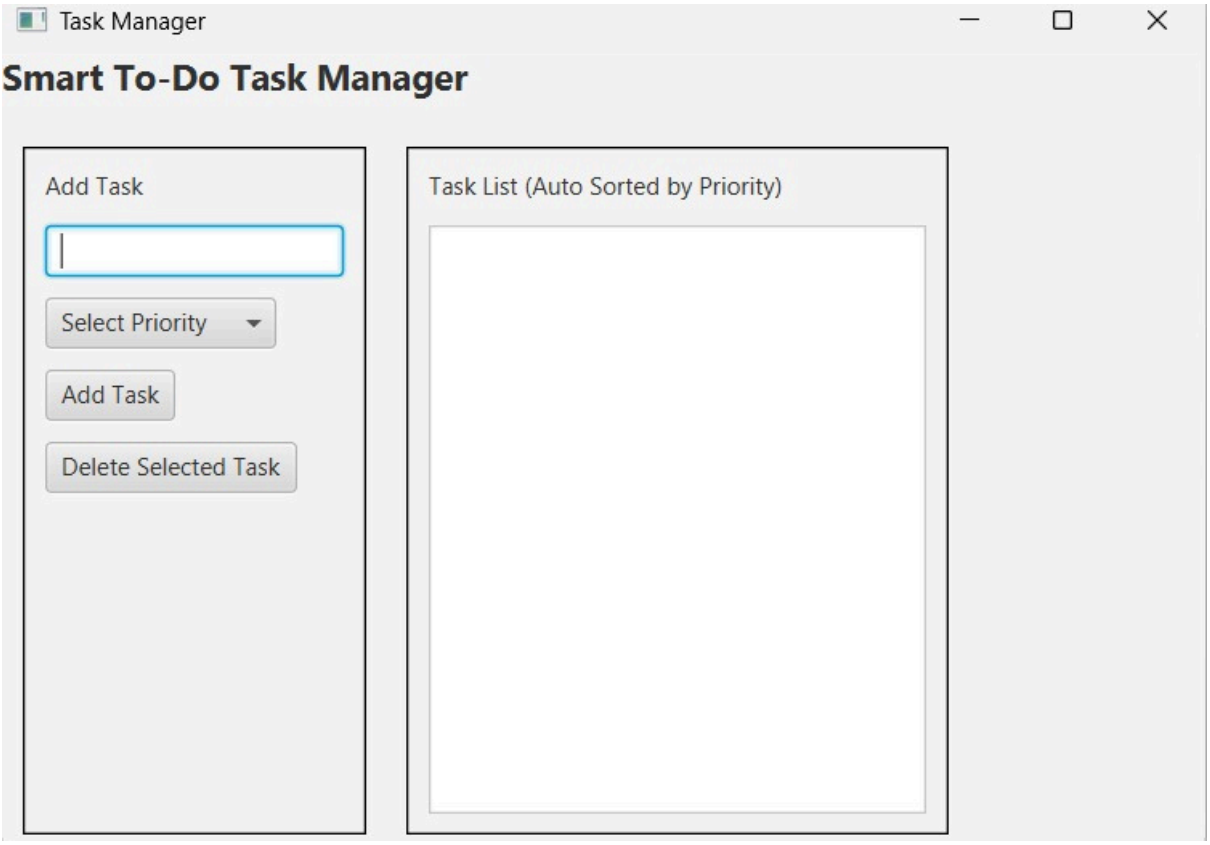
    VBox root = new VBox(10, title, mainLayout);

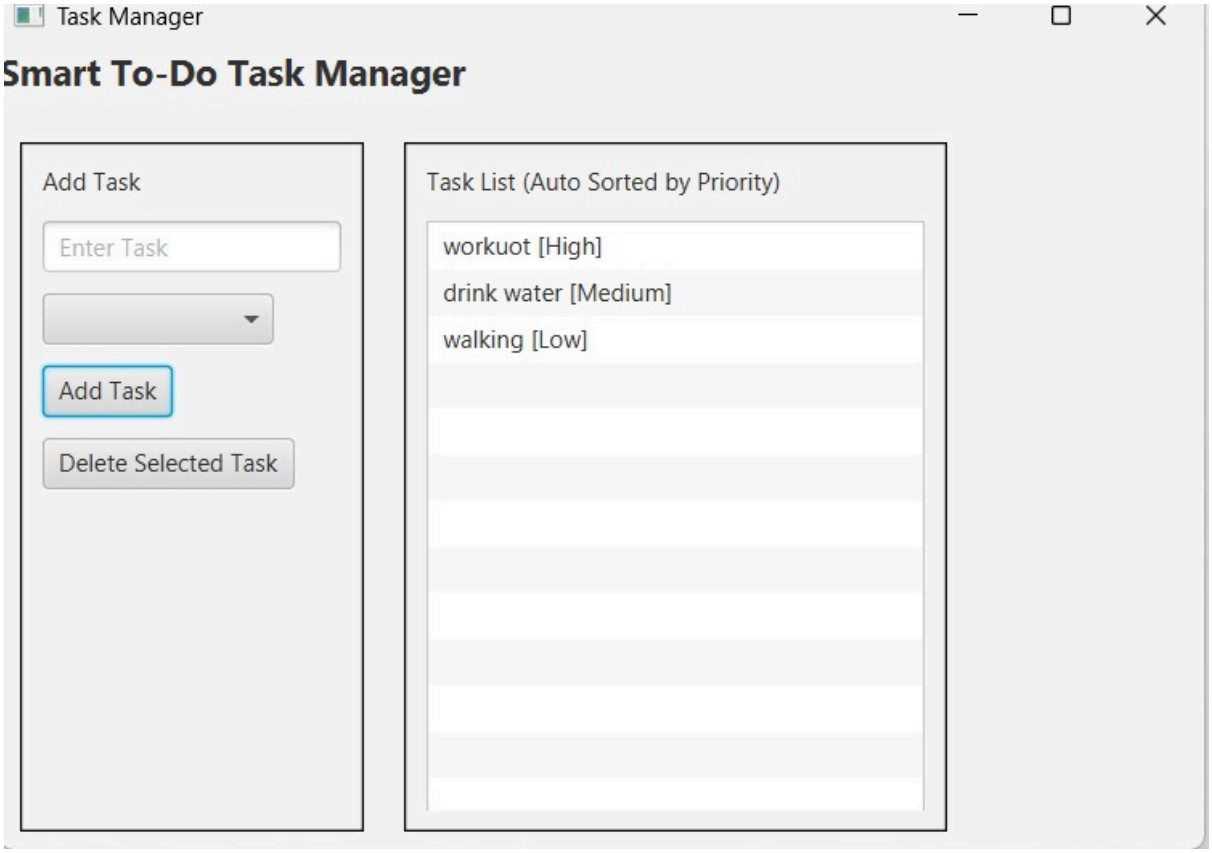
    Scene scene = new Scene(root, 600, 400);
    stage.setScene(scene);
    stage.setTitle("Task Manager");
    stage.show();
}

public static void main(String[] args) {
    launch();
```

```
    }
}
```

6. INPUT AND OUTPUT SCREENS




Task Manager
Smart To-Do Task Manager
Add Task
Enter Task
Add Task
Delete Selected Task
Task List (Auto Sorted by Priority)
workuot [High]
drink water [Medium]
walking [Low]

7. DISCUSSION & CONCLUSION

7.1 Discussion

The system was developed using JavaFX to automate hotel reservations. It successfully manages bookings, prevents double booking, and handles guest check-out efficiency

The system successfully manages tasks and prioritizes them automatically. It reduces manual effort and improves productivity.

7.2 Conclusion

The Smart To-Do Task Manager demonstrates how JavaFX can be used to build real-world applications. It improves task organization and helps users focus on important tasks.

8. REFERENCES

The following resources were used during the development of this project

Java Documentation (Oracle)

GeeksforGeeks

W3Schools